MOHAMMAD REHAN

Senior AI Systems Architect & Tech Lead

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EXECUTIVE SUMMARY

High-impact AI Systems Architect and Operations Manager with 4+ years of hands-on leadership bridging commercial retail operations and fault-tolerant software engineering. Architect of institutional-grade algorithmic trading systems with zero capital leakage, autonomous self-healing engines with sub-50ms MTTR, multi-modal generative AI pipelines, and commercial enterprise POS architectures.

CORE COMPETENCIES & ARCHITECTURE STACK

- **Distributed Systems & Fault Tolerance**: Write-Ahead Logging (WAL) Micro-Journaling, Continuous Chaos Engineering, Sub-50ms Process Supervision, Dynamic AST Poison Quarantine, Circuit Breakers.
- **Algorithmic Trading & FinTech**: Low-latency multi-agent derivatives execution (NIFTY/BANKNIFTY/SENSEX), Spot/15m VWAP structural stop-loss guards, 0-DTE theta decay shields, sub-8ms exchange proximity.
- **Generative AI & Media Engineering**: Autonomous LLM agent routing, Neural TTS voice synthesis, FFmpeg programmatic video composition, ASS animated subtitle engines.
- **Cloud Infrastructure & Security**: AWS Mumbai (ap-south-1), Cloudflare Pages Edge, Zero-Trust Network Access (ZTNA) via Private WireGuard Mesh, Linux systemd real-time priority tuning (Nice=-10).
- **Full-Stack & POS Development**: Python 3.12, AsyncIO, SQLite, Streamlit, HTML5/CSS3, Modern JavaScript, Thermal Printer Protocol, ReportLab PDF Engine.

KEY PRODUCTION SYSTEMS & ARCHITECTURES

Indian Quant Options Sniper -- Institutional Derivatives Execution Engine

- Engineered an institutional-grade, multi-agent algorithmic options execution engine deployed on AWS Mumbai (ap-south-1) with sub-8ms deterministic fiber proximity to NSE co-location.
- Implemented single direct atomic routing for <=24 lots (390 NIFTY qty), eliminating retail double-brokerage penalties and 500ms market-maker front-running lag.
- Anchored dynamic stop-loss to NIFTY Spot 15m VWAP structure rather than option premium noise, backed by a 35% emergency disaster ceiling and a 30% capital reserve lock.
- Automated a 13:30 IST Expiry Day Theta Filter to eliminate 0-DTE time decay slaughter, automatically routing breakout momentum to next-week contracts.
- Secured live operations under 24/7 systemd autopilot (0.00 load average, 2GB swap protection) isolated behind an encrypted private WireGuard mesh (ZTNA).

Project Phoenix -- Autonomous Fault-Tolerant Resilience Engine & 24/7 Chaos Sentinel

- Architected an antifragile self-healing framework enforcing the Golden Rule of Resilience: auto-healing into a permanently evolved, stronger state rather than vulnerable status-quo recovery.
- Implemented EvolutionaryHardener: automatic generation Epoch progression, SHA-256 perimeter signature quarantine, and +25% adaptive parameter stiffening.
- Certified sub-50ms MTTR (11ms-60ms) hot process resurrection with Ash Ledger write-ahead micro-journaling preventing zero state loss.
- Validated via automated Level-5 Doomsday Chaos Gauntlet: withstood 50-thread concurrent stampedes, 800MB heap shockwaves, ReDoS bombs, and disk sector corruption with 100.0% defense.
- Operates as a permanent 24/7 background sentinel running autonomous hourly resilience drills.

Autonomous AI Shorts & Reels Studio -- Multi-Modal Video Production Engine

- Developed an end-to-end automated vertical video (9:16) production studio combining topic-to-script LLM generation, studio neural voiceover, and dynamic B-roll synchronization.
- Engineered a custom ASS subtitle engine compiling word-by-word colored animated subtitle cues (Hormozi style) with automatic background audio ducking.
- Fortified rendering pipeline with Phoenix process supervision, preventing FFmpeg memory leaks and out-of-memory crashes during batch renders.
- Autonomously outputs ready-to-publish 1080x1920 Full HD video packages complete with high-CTR titles, SEO descriptions, and tags.

Footwear Retail POS & Executive Sales Dashboard

- Built a commercial Point of Sale (POS) and inventory ledger managing retail store operations, barcode scanning, thermal printing, and GST compliance.
- Features multi-counter concurrency protection, inventory reorder forecasting, and real-time sales margin dashboards.

Maison Ladies Zone -- Haute Couture Digital Showcase

- Architected a luxury digital showroom platform (ladieszonepatna.in) with appointment trial concierge workflows, client-side curator photo management, and a 5-layer mathematical stealth shield.
- Achieved Grade A+ 100/100 SEO optimization and sub-200ms global edge delivery on Cloudflare Pages.

PROFESSIONAL EXPERIENCE

Store Manager & Lead Systems Architect

Ladies Zone Exclusive Showroom -- Patna, Bihar, India *(March 2021 - Present)*

- Direct day-to-day operations, inventory replenishment, and customer relationship management for a high-volume retail fashion showroom.
- Architected and deployed in-house biometric attendance audit ledger, retail POS billing software, and custom business workflow automations.
- Bridged physical business requirements with automated software engineering to eliminate operational overhead and inventory discrepancies.

EDUCATION

Bachelor of Computer Applications (BCA)

National Business College -- Patna, Bihar, India *(Graduated January 2025)*

- Core coursework: Object-Oriented Software Design, Database Management Systems, Data Structures, Web Technologies, and Network Security.